Ewald Sum

The electrostatic energy:

$$U = \frac{1}{2}\sum_{i,j,n}{}' \frac{q_i q_j}{|r_i - r_j + nL|}$$

This is fine for short range, but for long range; this expression isn't computationally beneficial.
So, we do the long range part in the "k-space" (Fourier space) and the short range part in the "R-space" (Real Space)

Suppose we've N point charges @ $\{\underline{r}_1, \underline{r}_2, \ldots, \underline{r}_N\}$ of magnitudes $\{q_1, q_2, \ldots, q_N\}$

So, the spatial density: $\rho(\underline{r}) = \sum_j q_j \delta(\underline{r} - \underline{r}_j)$

The key idea is to break the density into an SR & an LR part.

$$\rho(\underline{r}) = \sum_j q_j \left[\delta(\underline{r} - \underline{r}_j) - \left(\frac{\alpha}{\pi}\right)^{3/2} \exp\left(-\alpha(\underline{r} - \underline{r}_j)^2\right)\right] + \sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp\left(-\alpha(\underline{r} - \underline{r}_j)^2\right)$$

SR part: $\sum_j q_j\left[\delta(\underline{r} - \underline{r}_j) - \left(\frac{\alpha}{\pi}\right)^{3/2}\exp\left(-\alpha(\underline{r} - \underline{r}_j)^2\right)\right]$: Point charge screened by a Gaussian Cloud

LR part: $\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp\left(-\alpha(\underline{r} - \underline{r}_j)^2\right)$: Only the Gaussian Cloud, which has a longer range.

LR Part:

(From the traditional convention of taking $\frac{1}{4\pi\epsilon_0} \sim 1$)

$$\int \underline{E}\cdot d\underline{S} = 4\pi Q \Rightarrow \int (\underline{\nabla}\cdot\underline{E})\, dV = 4\pi \int \rho\, dV$$

$$\Rightarrow \underline{\nabla}\cdot(-\underline{\nabla}\phi) = 4\pi\rho \Rightarrow \boxed{\nabla^2\phi = -4\pi\rho}$$

F.T. of this eq^n: $k^2 \tilde{\phi}(k) = 4\pi \tilde{\rho}(\underline{k})$ or $k^2 \tilde{\phi}_{long}(\underline{k}) = 4\pi \tilde{\rho}_{long}(\underline{k})$

Also, $\rho_{long}(\underline{r}) = \sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp\left(-\alpha(\underline{r}-\underline{r}_j)^2\right)$

defining $FT[f(\underline{x})] = \frac{1}{V}\int_{x\text{-space}} f(\underline{x}) \exp(-i\underline{k}\cdot\underline{x})\, d\underline{x}$

F.T. of $\rho_{long}(\underline{r})$: $\tilde{\rho}_{long}(\underline{k}) = \frac{1}{V}\int \sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp\left(-\alpha(\underline{r}-\underline{r}_j)^2\right)\exp(-i\underline{k}\cdot\underline{r})\, d\underline{r}$

$\Rightarrow \tilde{\rho}_{long}(\underline{k}) = \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \int \exp\left(-\alpha(\underline{r}-\underline{r}_j)^2\right)\exp(-i\underline{k}\cdot\underline{r})\, d\underline{r}$

$\underline{r}-\underline{r}_j = \underline{t}$

$\tilde{\rho}_{long}(\underline{k}) = \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \int \exp(-\alpha t^2)\exp\left(-i\underline{k}\cdot(\underline{t}+\underline{r}_j)\right) d\underline{t}$

$= \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp(-i\underline{k}\cdot\underline{r}_j) \int \underbrace{\exp(-\alpha t^2 - i\underline{k}\cdot\underline{t})\, d\underline{t}}_{\text{Break into components of } t}$

$\tilde{\rho}_{long}(\underline{k}) = \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp(-i\underline{k}\cdot\underline{r}_j)\left(\int \exp(-\alpha X^2 - ik_x X)\, dX\right)\left(\underset{y\text{-part}}{\;}\right)\left(\underset{z\text{-part}}{\;}\right)$

$\tilde{\rho}_{long}(\underline{k}) = \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp(-i\underline{k}\cdot\underline{r}_j)\left(\int \exp\left(-\alpha\left(X^2 + 2X\frac{ik_x}{2\alpha} - \frac{k_x^2}{4\alpha^2} + \frac{k_x^2}{4\alpha^2}\right)\right)dX\right)(\;)(\;)$

$\Rightarrow \tilde{\rho}_{long}(\underline{k}) = \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp(-i\underline{k}\cdot\underline{r}_j)\left(\int \exp\left(-\alpha\left(X+\frac{ik_x}{2\alpha}\right)^2 - \frac{k_x^2}{4\alpha}\right)dX\right)(\;)(\;)$

$= \frac{1}{V}\sum_j q_j \left(\frac{\alpha}{\pi}\right)^{3/2} \exp(-i\underline{k}\cdot\underline{r}_j)\left(\exp\left(-\frac{k_x^2}{4\alpha}\right)\sqrt{\frac{\pi}{\alpha}}\right)\underbrace{\left(\exp\left(-\frac{k_y^2}{4\alpha}\right)\sqrt{\frac{\pi}{\alpha}}\right)}_{y\text{-part}}\underbrace{\left(\exp\left(-\frac{k_z^2}{4\alpha}\right)\sqrt{\frac{\pi}{\alpha}}\right)}_{z\text{-part}}$

$\Rightarrow \frac{1}{V}\sum_j q_j \exp(-i\underline{k}\cdot\underline{r}_j)\exp(-k^2/4\alpha) = \tilde{\rho}_{long}(\underline{k})$

And: $\tilde{\phi}_{long}(\underline{k}) = \frac{4\pi}{k^2}\tilde{\rho}_{long}(\underline{k})$

$$\Rightarrow \tilde{\phi}_{long}(\underline{k}) = \frac{4\pi}{k^2 V} \sum_j q_j \exp(-i\underline{k}\cdot\underline{r}_j) \exp\left(-\frac{k^2}{4\alpha}\right)$$

Now, we've $\tilde{\phi}_{long}(\underline{k})$; we can find U_{long}; if we can invert it. $\tilde{\phi}_{long}(\underline{k})$

for inv. FT, $FT^{-1}[F(\underline{k})] = \left(\frac{1}{2\pi}\right)^3 \times V \times \int_{\underline{k}} F(\underline{k}) \exp(i\underline{k}\cdot\underline{r})\, d\underline{k}$

In Discrete space:

$$FT^{-1}[F(\underline{k})] = \frac{V}{(2\pi)^3} \sum_{\underline{k}} F(\underline{k}) \exp(i\underline{k}\cdot\underline{r}) \underbrace{\left(\frac{2\pi}{L}\right)}_{\Delta k_x} \underbrace{\left(\frac{2\pi}{L}\right)}_{\Delta k_y} \underbrace{\left(\frac{2\pi}{L}\right)}_{\Delta k_z}$$

$$FT^{-1}[F(\underline{k})] = \sum_{\underline{k}} F(\underline{k}) \exp(i\underline{k}\cdot\underline{r})$$

$$\Rightarrow \phi_{long}(\underline{r}) = \sum_{\underline{k}} \frac{4\pi}{k^2 V} \sum_j q_j \exp(-i\underline{k}\cdot\underline{r}_j) \exp(i\underline{k}\cdot\underline{r}) \exp\left(-\frac{k^2}{4\alpha}\right)$$

$$\phi_{long}(\underline{r}) = \sum_{j,\underline{k}} \frac{4\pi q_j}{k^2 V} \exp\left(i\underline{k}\cdot(\underline{r}-\underline{r}_j)\right) \exp\left(-\frac{k^2}{4\alpha}\right)$$

$$\& \; U_{long} = \frac{1}{2} \sum_i \phi_{long}(\underline{r}_i) q_i = \frac{1}{2} \sum_{i,j,\underline{k}} \frac{4\pi q_i q_j}{k^2 V} \exp\left(i\underline{k}\cdot(\underline{r}_i-\underline{r}_j)\right) \exp\left(-\frac{k^2}{4\alpha}\right)$$

$$\tilde{\rho}(\underline{k}) = FT\left[\sum_j q_j \delta(\underline{r}-\underline{r}_j)\right] = \frac{1}{V} \sum_j q_j \exp(-i\underline{k}\cdot\underline{r}_j)$$

(Not long or short)

$$\tilde{\rho}^*(\underline{k}) = \frac{1}{V} \sum_i q_i \exp(-i\underline{k}\cdot\underline{r}_i)$$

$$\therefore U_{long} = \frac{1}{2}\sum_{i,j,k} \frac{4\pi q_i q_j}{k^2 V} \exp\left(i\underline{k}\cdot(\underline{r}_i - r_j)\right)\exp\left(\frac{-k^2}{4\alpha}\right)$$

$$= \frac{1}{2}\sum_{i,j,k} V^2 \times \frac{4\pi}{k^2 V} |\tilde{\rho}(\underline{k})|^2 \exp\left(\frac{-k^2}{4\alpha}\right)$$

$$U_{long} = \frac{1}{2}\sum_{i,j,k} \frac{4\pi V}{k^2} |\tilde{\rho}(\underline{k})|^2 \exp(-k^2/4\alpha)$$

← The LR eg is decaying exponentially! Not as $1/r$; $\exp(-k^2/4\alpha)$ is a much faster decay!!

So we can easily compute U_{long} by considering only a (relatively) fewer pts in the k-space.

But;

we also need to remove the interaction of the cloud w ~~itself~~ a charge @ the center. (which crept in the U_{long} expr.) : The self energy.

Assume both cloud + the pt. charge are @ the origin! (Simplifies the analysis)

$$-\nabla^2\phi^{Gauss} = 4\pi\rho^{Gauss} ; \quad \rho^{Gauss} = q_i\left(\frac{\alpha}{\pi}\right)^{3/2}\exp(-\alpha r^2)$$

$$\phi^{Gauss} = q_i \operatorname{erf}(r\sqrt{\alpha})$$

(Assuming $r\phi^{Gauss} \to 0$ as $r\to\infty$ & $\phi^{Gauss} \to 0$ as $r\to\infty$)

Since we're interested near $r=0$; we'll do a T-Series expansion

near $r=0$: $\phi^{Gauss}(r=0) = 2q_i\left(\frac{\alpha}{\pi}\right)^{1/2}$ ← Had we assumed the cloud + charge centered @ r_j: we would've found around $r=r_j$

$$\phi^{Gauss} = 2q_i\left(\alpha/\pi\right)^{1/2}$$

$$U^{Gauss} = \frac{1}{2} q_i \phi^{Gauss}$$ Std. formula for self i/a eg

$$\Rightarrow U^{Gauss} = q_i^2\left(\frac{\alpha}{\pi}\right)^{1/2}$$

For all charges: $$U^{Gauss} = \left(\frac{\alpha}{\pi}\right)^{1/2}\sum_i q_i^2$$

So:

$$U_{long}^{correct} = \frac{1}{2}\sum_{i,j,k} \frac{4\pi V}{k^2} |\tilde{\rho}(\underline{k})|^2 \exp\left(\frac{-k^2}{4\alpha}\right) - \left(\frac{\alpha}{\pi}\right)^{1/2} \sum_i q_i^2$$

<u>SR Part</u>

$\phi_{short}(r)$: Can be obtained by solving Poisson eq^n

$$\phi_{short}(r) = \frac{q_i}{r} - \frac{q_i}{r} \operatorname{erf}(\sqrt{\alpha}\, r) = \frac{q_i}{r} \operatorname{erfc}(\sqrt{\alpha}\, r)$$

$$U_{short} = \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} \frac{q_i q_j}{r_{ij}} \operatorname{erfc}(r_{ij}\sqrt{\alpha})$$

To prevent self

Finally:

$$U = U_{long}^{correct} + U_{short}$$

$$U = \left(\frac{1}{2}\sum_{i,j,k} \frac{4\pi V}{k^2} |\tilde{\rho}(\underline{k})|^2 \exp\left(\frac{-k^2}{4\alpha}\right) - \left(\frac{\alpha}{\pi}\right)^{1/2} \sum_i q_i^2\right) + \left(\frac{1}{2}\sum_{\substack{i,j \\ i \neq j}} \frac{q_i q_j}{r_{ij}} \operatorname{erfc}(r_{ij}\sqrt{\alpha})\right)$$